

Tran Nguyen Viet Hoang

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SUMMARY

Third-year Computer Networks and Data Communications student pursuing a **DevOps Intern** role. Interested in containerization, infrastructure, cloud-native platforms, CI/CD pipeline and MLOps. Architected scalable systems using Docker, Kubernetes, Terraform, and GitHub Actions.

EDUCATION

University of Information Technology, VNU-HCM

Ho Chi Minh City, Vietnam

B.Eng. in Computer Networks and Data Communications

2023 – Expected 2027

– GPA: 8.71/10.00

PROJECTS

MLOps PaaS System | GitHub | *Django, FastAPI, Redpanda, PostgreSQL, Docker, Kubernetes, AWS, GitHub Actions, ArgoCD, Argo Workflows, MLflow, KubeFlow, BentoML* 3/2026 – Present

- Built a multi-tenant MLOps platform automating model training, dynamic deployment, version management, and data drift monitoring for diverse ML/DL models (XGBoost, Scikit-Learn, PyTorch, TensorFlow).
- Developed a Django RESTful API as the central control system for managing user resources, model state, and an immutable registry, with Argo Workflows orchestrating all Kubernetes workloads (build, deploy, train, drift).
- Packaged multi-framework models via MLflow Model Packaging, then dynamically served ML models through FastAPI + MLflow Model Loading and DL models through BentoML, each deployed as isolated per-version pods.
- Implemented data drift monitoring using Evidently AI, comparing reference datasets against live production data streamed through a Redpanda event pipeline and persisted by a KEDA-scaled consumer.
- Orchestrated distributed model training with KubeFlow Training Operator (PyTorchJob) and dynamically provisioned GPU/CPU compute nodes on-demand via Karpenter auto-scaling.
- Achieved zero-downtime model serving using rolling updates and HPA, benchmarked at ~100 concurrent users with p95 latency ≤ 100 ms and throughput of ≥ 50 req/s.
- Implemented a GitOps-driven CI/CD pipeline using GitHub Actions and ArgoCD, automating code quality checks, unit testing, container image scanning, building, and declarative cluster state synchronization via Kustomize.

SageLMS | GitHub | *Spring Boot, FastAPI, LangChain, Redis, PostgreSQL, Docker, Kubernetes, GCP, GitHub Actions, FluxCD, Kyverno, Grafana & Prometheus* 4/2026 – 6/2026

- Architected a secure-by-design microservices platform (Spring Boot gateway, auth, course, assessment, challenge, and a FastAPI + LangChain AI tutor) following DevSecOps principles across the entire software lifecycle.
- Implemented a shift-left CI gate on every PR: Conventional Commit linting, secret scanning (Gitleaks), path-filtered unit tests (JUnit/pytest/Vitest), Dockerfile linting (Hadolint), dependency vulnerability scanning (Trivy filesystem), IaC validation (OpenTofu fmt/validate + Checkov), and static code analysis with quality gate enforcement (SonarCloud).
- Built an automated CD pipeline that detects changed services, builds Docker images, Trivy scans for HIGH/CRITICAL vulnerabilities, pushes to a private Harbor registry, then cryptographically signs each image and attests its CycloneDX SBOM using Cosign, and finally opens a deploy PR that pins image digests in the Kustomization manifest for FluxCD to reconcile.
- Enforced supply-chain and runtime security in the Kubernetes cluster via Kyverno admission policies: requiring images from the trusted Harbor registry only, pinning by immutable SHA-256 digest, disallowing the mutable `:latest` tag, verifying Cosign keyless signatures against the CI workflow identity, enforcing Pod Security Baseline, requiring resource limits/health probes, and mandating non-root containers with no privilege escalation.
- Managed secrets securely through GCP Workload Identity Federation for passwordless CI/CD authentication, External Secrets Operator for runtime credential injection, and automated post-deploy verification (GKE rollout status checks and endpoint health probes).

RESEARCH

CSoNet 2026: A Drift-Aware MLOps Framework for Multi-Model Serving: A Network Intrusion Detection Case Study

Preview

- First author; proposed a cloud-native MLOps architecture for multi-model ML and DL lifecycle management, featuring a decoupled control-plane/data-plane design to separate model governance from online prediction serving.
- Addressed the data drift problem—where changing production data distributions degrade model reliability—by designing a monitoring pipeline that asynchronously logs predictions, computes statistical drift scores over production windows, and triggers review-driven adaptation actions (e.g., alerts, retraining).
- Validated the framework on a CIC-IDS2017 Network Intrusion Detection case study across 3 attack families $\times$ 3 severity levels; drift-triggered retraining recovered Macro-F1 from 0.13–0.46 (fixed model) to 0.98–1.00 (retrained challenger), demonstrating Δ Macro-F1 gains of +0.54 to +0.87.

AWARDS

Encouragement Scholarship for Academic Excellence

- Semester 1, 2023–2024
- Semester 1, 2024–2025
- Semester 2, 2024–2025

TECHNICAL SKILLS

Software: ReactJS, Django, FastAPI, PostgreSQL, Redis, Redpanda/Kafka

Containers & Orchestration: Docker, Kubernetes, EKS, GKE, Argo Workflows

Infrastructure: GCP, AWS, Terraform, Ansible, Helm

CI/CD & GitOps: Git, GitHub, GitHub Actions, Argo CD, Flux CD

MLOps: MLflow, Kubeflow, BentoML

Observability & Networking: Prometheus, Grafana, Cloudflare, Cisco Packet Tracer